

Date	30 June 2026	Maximum Marks	2 Marks
Team ID		Project Name	Strategic Product Placement Analysis System

Project Design Phase

Proposed Solution Template

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template to clearly define the value proposition and operational model of the project.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Physical retail lacks granular, automated spatial analytics for product placements. Merchandisers struggle to translate raw 2D POS spreadsheet data into optimal physical 3D display strategies without relying on expensive hardware or guesswork, leading to sub-optimal cross-merchandising and lost revenue.
2.	Idea / Solution description	A hardware-free, AI-driven spatial mapping web platform. It ingests standard store floor plans (PDF/ PNG) and POS sales data (CSV) to generate interactive heatmaps. The system predicts impulse-buy hotspots and recommends optimal aisle placements for secondary displays.
3.	Novelty / Uniqueness	Unlike existing solutions that require massive capital expenditure and logistical hurdles (e.g., installing IoT shelf sensors or smart-tracking cameras), this system leverages existing historical POS data and standard architectural floor plans to provide a purely software-based, low-cost spatial analytics engine.
4.	Social Impact / Customer Satisfaction	• For Retailers: Empowers store managers with data-driven confidence, reducing the frustration of "trial and error" merchandising and proving ROI to brand partners. • For Shoppers: Enhances the shopping experience by presenting complementary products logically across the store, reducing navigation fatigue.

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5.	Business Model (Revenue Model)	B2B SaaS (Software as a Service) subscription model. Pricing is tiered based on the number of retail locations analyzed or the volume of POS data processed monthly. Includes potential one-off onboarding fees for initial floor-plan digitization.
6.	Scalability of the Solution	Highly scalable due to its cloud-native, containerized microservices architecture (Docker/Kubernetes). Because it requires zero hardware installation, the platform can seamlessly scale from analyzing a single boutique store to an entire nationwide retail chain instantly.